



Aishwarya Bhargava

Senior Machine Learning Engineer (5+ years experience)

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Profiles

[in aishwaryabhargava](#)

[aish2997](#)

Skills

Machine Learning & AI



Statistics, Exploratory Data Analysis (EDA), Numpy, Pandas, Scikit-learn, Matplotlib, Seaborn, Machine Learning, Deep Learning, TensorFlow, PyTorch, Jupyter Notebooks, OpenCV, Computer Vision, Object Detection, Image Segmentation, GenerativeAI, Transformers, LLaMAIndex, LangChain, CNN, Prompt Engineering

DevOps and Cloud Engineering



Git, GitHub, GitHub Actions, GitHub Workflows, Terraform, Docker, Kubernetes, GCP Cloud Functions, GCP Cloud Run, GCP Cloud Scheduler, GCP Cloud Workflows, GCP Vertex AI, Nginx, Unicorn, Load Balancers, Compute Engine, Linux, CI/CD

Database & Data Engineering



SQL, BigQuery, dbt (data build tool), Google Cloud Platform (GCP), Google Cloud Storage (GCS), Pub/Sub

Software Engineering & Agile



Python, FastAPI, Flask, Agile, Scrum

Soft Skills



Communication, Collaboration, Problem-Solving, Critical Thinking, Adaptability, Time Management, Result-Oriented

Education

Guru Gobind Singh Indraprastha University

August 2015 - May 2019

Bachelors of Technology (IT)

Certifications

Associate Cloud Engineer

Google Cloud

Professional Data Engineer

Google Cloud

Deep Learning Specialization

DeepLearning.AI

DeepLearning.AI - Generative AI with Large Language Models

DeepLearning.ai

Summary

Senior Machine Learning Engineer with extensive experience in driving data-driven solutions, developing AI and Data Engineering systems, and streamlining deployment processes. Demonstrated expertise in executing end-to-end deep learning projects, optimizing data pipelines, and mentoring teams to enhance solution quality.

Experience

Tata Consultancy Services

Dec 2022 - Present

AI Consultant at Hennes & Mauritz Retail Pvt. Ltd.

Stockholm, Sweden

- Collaborated with business stakeholders to understand challenges, define solution requirements, and identify relevant data sources. Partnered with cross-functional teams to gather necessary data for model development.
- Contributed to the complete machine learning workflow, including data gathering, exploratory data analysis, feature engineering, feature selection, model building/training, optimization, and deployment.
- Managed the end-to-end ML project lifecycle, from dataset preparation to CI/CD pipeline implementation, within an Agile team environment.
- Developed Knowledge Retrieval Systems, leveraging advanced Retrieval-Augmented Generation (RAG) techniques and integrating Large Language Models (LLMs) to deliver enhanced contextual responses.
- Designed and deployed custom Cloud Functions to automate API data ingestion and decrypt encrypted files into BigQuery.
- Automated data orchestration and storage using Cloud Schedulers, Cloud Workflows, and Google Cloud Storage (GCS).
- Streamlined diverse data ingestion workflows, integrating data from APIs, Google Analytics, GCS, and Azure Blob into BigQuery.

Tata Consultancy Services

Aug 2021 - Nov 2022

Senior Machine Learning Engineer at Johnson & Johnson Inc.

Delhi, India

- Trained an image segmentation model on a custom-annotated dataset for precise detection of surgical equipment welds.
- Applied advanced post-processing techniques to segmentation masks to improve weld detection accuracy and integrated mathematical logic to identify different weld defects.
- Deployed models for robust production performance, achieving a 98% reduction in inspection time.

Tata Consultancy Services

July 2019 - July 2021

Machine Learning Engineer at Ministry of Education, Govt. of India

Delhi, India

- Developed and optimized computer vision models in areas such as image classification, object detection, image segmentation, and facial authentication, leveraging models like EfficientNet, Detectron2, and Yolact++.
- Contributed to all stages of the machine learning project lifecycle, from dataset preparation and model training to improving accuracy and deploying models on GCP. This experience ensures successful production deployment and performance optimization.
- Customized models for client-specific requirements by retraining them on custom-annotated datasets.
- Applied optimization techniques such as quantization to address computational constraints and cythonization to minimize inference times.
- Achieved a 95% model performance rate in production, processing 400,000 images daily within 6 hours.